Galaba Vamsi

Software Engineer | AI/ML | Computer Vision | Systems
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PROFESSIONAL SUMMARY

Mechatronics undergraduate at IIT Bhilai focused on software engineering, AI/ML, computer vision, robotics, and GPU systems. Builds full-stack products, open-source tooling, simulators, and visual applications with Python, TypeScript, React, PyTorch, CUDA, and cloud platforms.

SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Kotlin, Swift, Dart, Go, GLSL, CUDA.
AI/ML: PyTorch, Graph Neural Networks (GATv2), Safe Reinforcement Learning, Transformers, OpenCV, CuPy.
Software and Systems: React, React Native, Flutter, Node.js, PostgreSQL, AWS, Supabase, Git, Linux, GitHub Actions, CI/CD, ROS 2, MEEP, OpenMPI, SDR/USRP.

EXPERIENCE

Open-Source Contributor - HFlow 2026
Hebbian Robotics (YC S26) | HFlow repository
- Contributed three maintainer-reviewed and merged pull requests to an open-source SDK for scalable multimodal data pipelines in robotics and Physical AI.
- PR #350 / issue #296: added paginated LeRobot tree traversal with `Link: rel="next"`, preserved request headers, deduplicated paths, rejected unsafe cross-origin URLs, and prevented pagination loops.
- PR #353 / issue #292: fixed LeRobot video-cache collisions by including camera, chunk, and file indexes in cache identity and bumping the converter version.
- PR #360 / issue #314: centralized safe ingest URI parsing across the CLI, server, and SDK, including validation for blank, absolute, and parent-escaping paths.

Research Intern - 6G RIS Simulator and Testbed Development May 2025 - Aug 2025
IIT Bhilai | Advisor: Dr. Arzad Alam
- Built a zero-budget, open-source Python simulator with USRP-like interfaces for line-of-sight and non-line-of-sight wireless environments.
- Designed and validated a two-layer reconfigurable intelligent surface (RIS) PCB workflow for self-powered RF energy harvesting; produced related first-author and co-author research.

PROJECTS

Human Slop - Co-Founder 2024 - 2026
Built a deployed anti-AI web and Android platform with React, PostgreSQL, and AWS; implemented hardware-bound biometric authentication and real-time typing forensics. [Website] | [App]

Arista Networks Wi-Fi Optimization System - Inter IIT 5th Rank 2025
Built a Python simulator and AI-assisted radio resource management framework using Dual-CNNs, Graph Neural Networks, and Safe Reinforcement Learning.

Electron-GNN - ML-Accelerated Quantum Spectroscopy 2025 - Present
Built a two-tower GATv2 model to predict molecular absorption spectra from geometry, targeting expensive RT-TDDFT simulation workflows. [GitHub]

Open-PyFX - GPU-Accelerated Video Effects 2025
Built a Python and GLSL video effects suite with GPU rendering for CRT emulation, ASCII edge detection, retro graphics, and 3D LUT pipelines. [GitHub]

Lunar Crater Detection - DEM Terrain Analysis 2025
Built a computer vision pipeline for crater extraction from Digital Elevation Models, achieving F1 score 0.839 and approximately 4x preprocessing speedup with CuPy/CUDA. [GitHub]

AntennaNet - Inverse Electromagnetic Design 2025
Developed KD-tree nearest-neighbour spectral anchoring for inverse electromagnetic design across 144D search spaces using Python, MEEP, and OpenMPI. [GitHub]

Amazon ML Challenge - Top 3.5% 2025
Led multimodal price prediction using Qwen 2-VL 7B and a ViT/RoBERTa two-tower model over 75,000 data pairs; achieved SMAPE 49.0.

EDUCATION

B.Tech. in Mechatronics Aug 2023 - May 2027
Indian Institute of Technology Bhilai
Coursework: Data Structures and Algorithms, Object-Oriented Programming, Deep Learning for Computer Vision, AI and Machine Learning, Digital Image Processing, Optimal Control.

PUBLICATIONS

First author: *An Open Emulator for Smart Radio Environments*. [Link]
Co-author: *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025. [Link]
Co-author: *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*. [Link]

ACHIEVEMENTS

1st Place: IIT Bhilai Web Development Competition. **5th Rank:** Arista Networks Wi-Fi Challenge, Inter IIT Tech Meet 14.0.
Top 3.5%: Amazon ML Challenge. **YC Startup School:** Accepted with Human Slop. **Top 10 Finalist:** Toyota Hackathon.